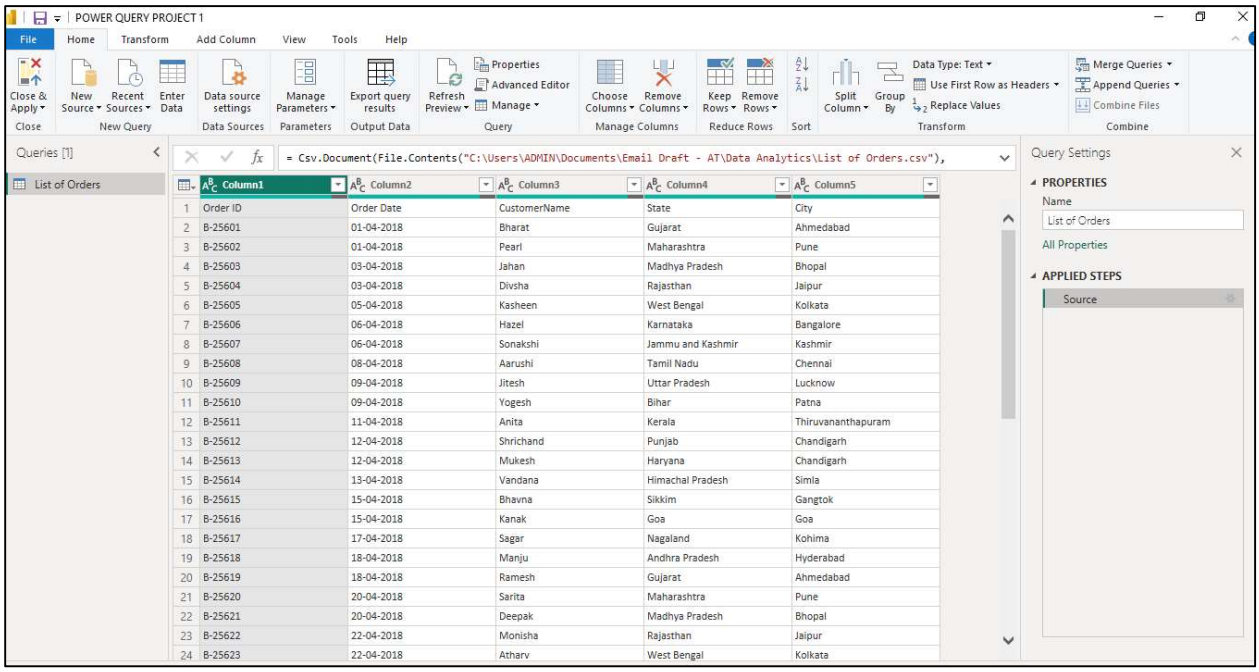
POWER QUERY PROJECT 1 – SOLUTIONS:

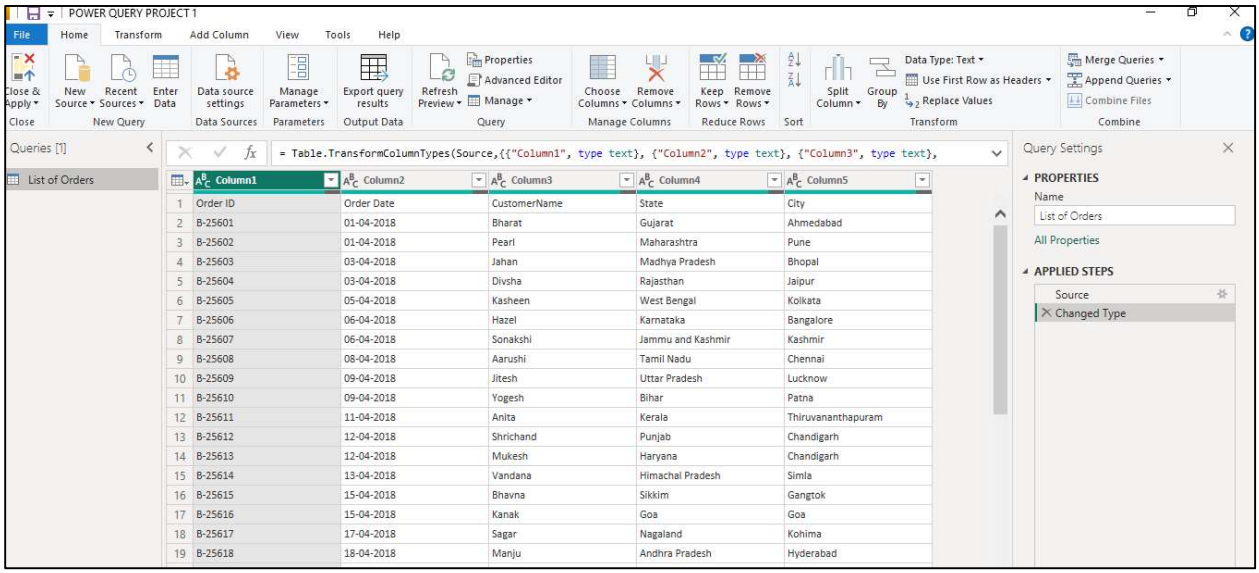
1. Data Import:

○ Import List of Orders.csv into Power BI.



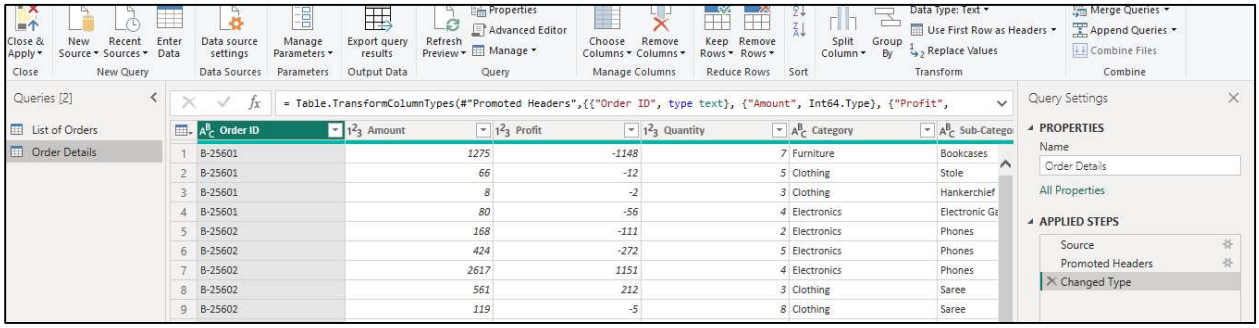
	Column1	Column2	Column3	Column4	Column5
1	Order ID	Order Date	CustomerName	State	City
2	B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
3	B-25602	01-04-2018	Pearl	Maharashtra	Pune
4	B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
5	B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
6	B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
7	B-25606	06-04-2018	Hazel	Karnataka	Bangalore
8	B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
9	B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai
10	B-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
11	B-25610	09-04-2018	Yogesh	Bihar	Patna
12	B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram
13	B-25612	12-04-2018	Shrichand	Punjab	Chandigarh
14	B-25613	12-04-2018	Mukesh	Haryana	Chandigarh
15	B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla
16	B-25615	15-04-2018	Bhavna	Sikkim	Gangtok
17	B-25616	15-04-2018	Kanak	Goa	Goa
18	B-25617	17-04-2018	Sagar	Nagaland	Kohima
19	B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad
20	B-25619	18-04-2018	Ramesh	Gujarat	Ahmedabad
21	B-25620	20-04-2018	Sarita	Maharashtra	Pune
22	B-25621	20-04-2018	Deepak	Madhya Pradesh	Bhopal
23	B-25622	22-04-2018	Monisha	Rajasthan	Jaipur
24	B-25623	22-04-2018	Atharv	West Bengal	Kolkata

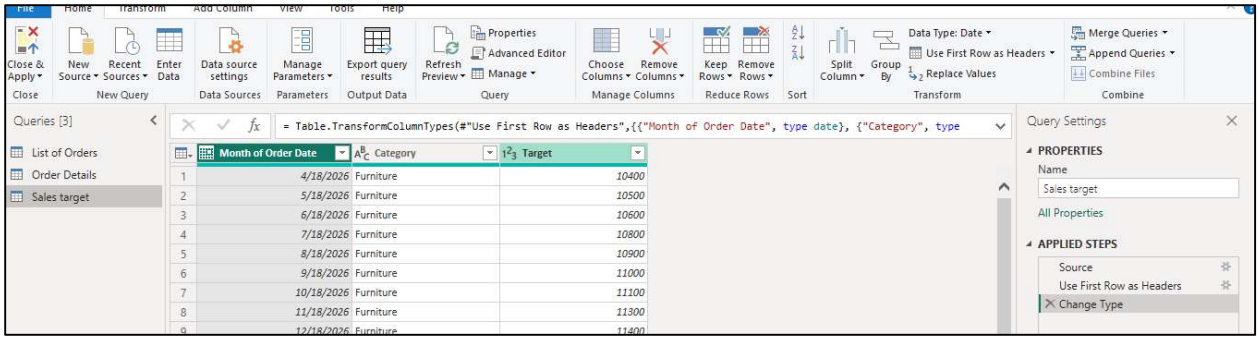
- Open the list of orders in the Power Query Editor using the Transform Data option.



	Column1	Column2	Column3	Column4	Column5
1	Order ID	Order Date	CustomerName	State	City
2	B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
3	B-25602	01-04-2018	Pearl	Maharashtra	Pune
4	B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
5	B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
6	B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
7	B-25606	06-04-2018	Hazel	Karnataka	Bangalore
8	B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
9	B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai
10	B-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
11	B-25610	09-04-2018	Yogesh	Bihar	Patna
12	B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram
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15	B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla
16	B-25615	15-04-2018	Bhavna	Sikkim	Gangtok
17	B-25616	15-04-2018	Kanak	Goa	Goa
18	B-25617	17-04-2018	Sagar	Nagaland	Kohima
19	B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad

- Import Order Details.csv and Sales Target.csv into Power Query Editor.

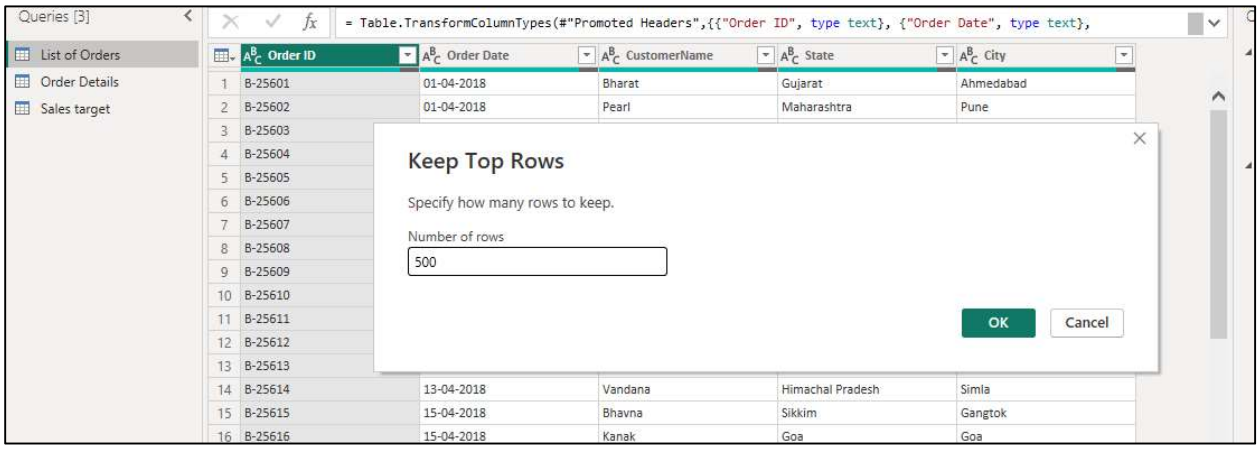




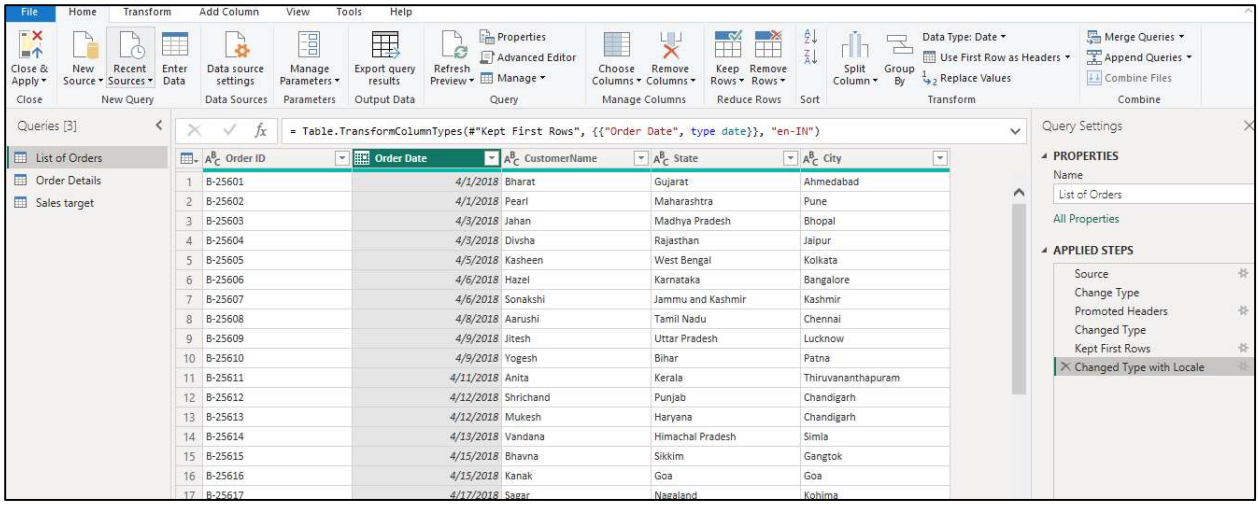
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2. Row Limiting & Data Types:

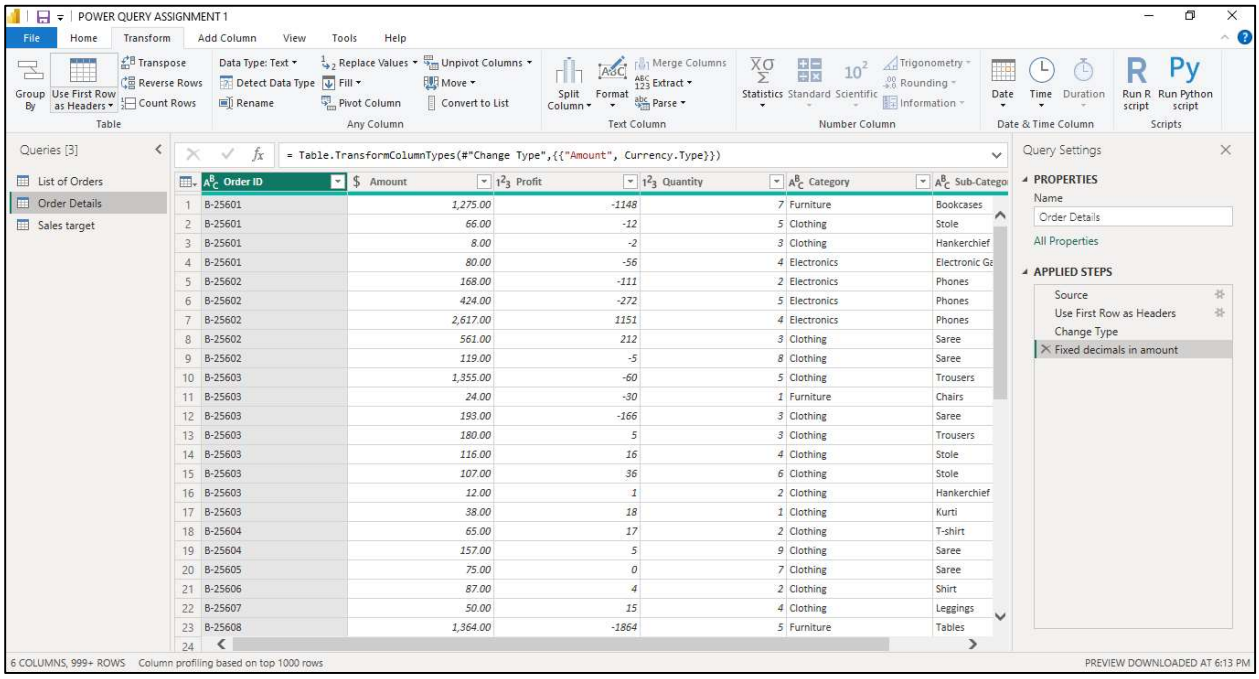
○ Restrict the List of Orders table to the first 500 rows.



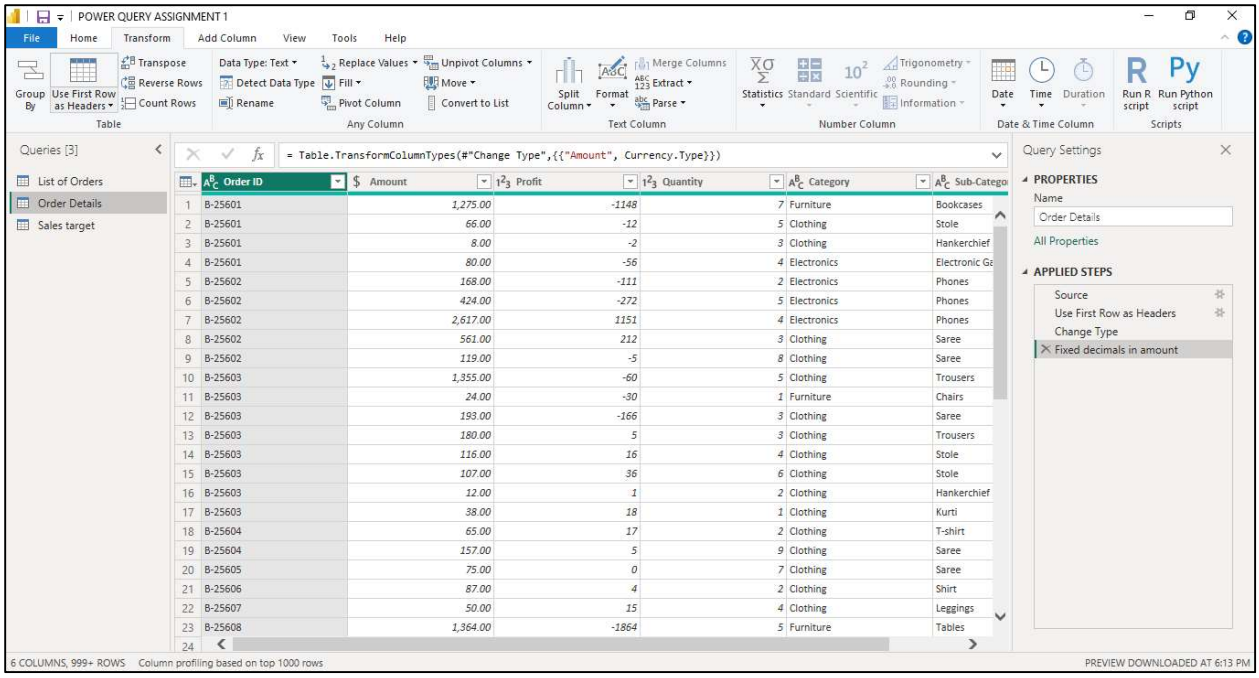
- Set the Order Date column to the Date data type.



Change the Amount and Target columns to fixed decimal numbers.

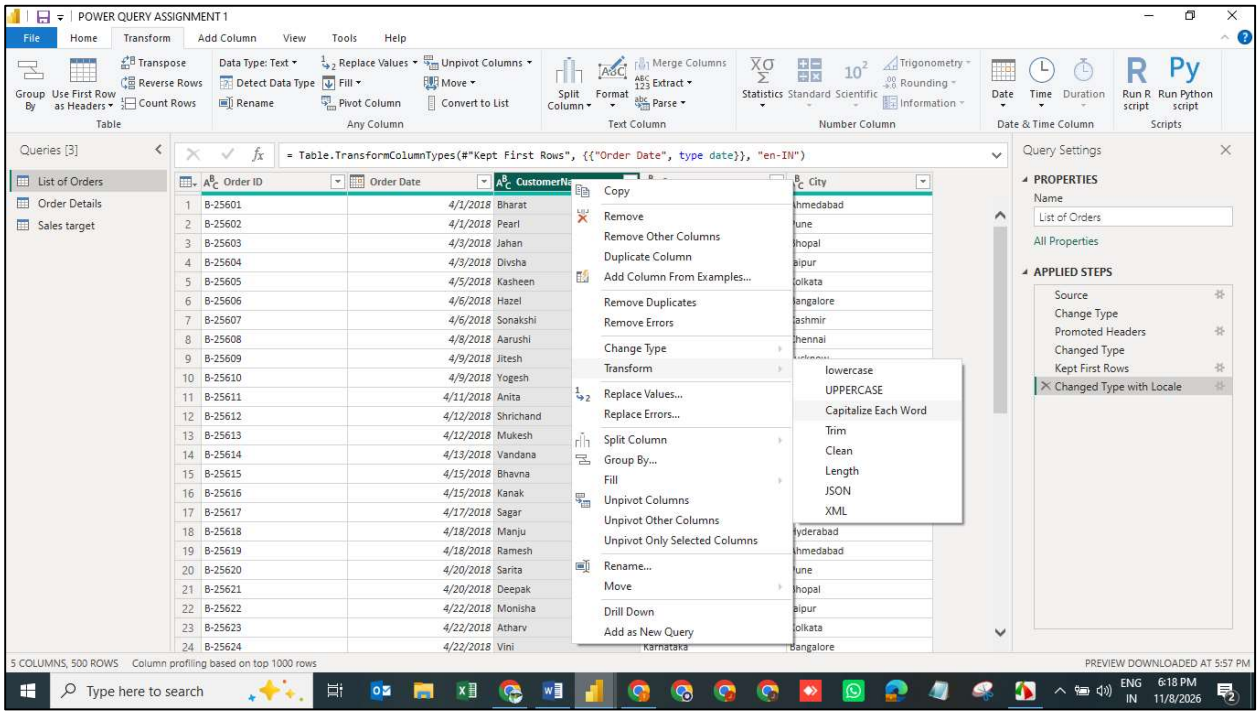


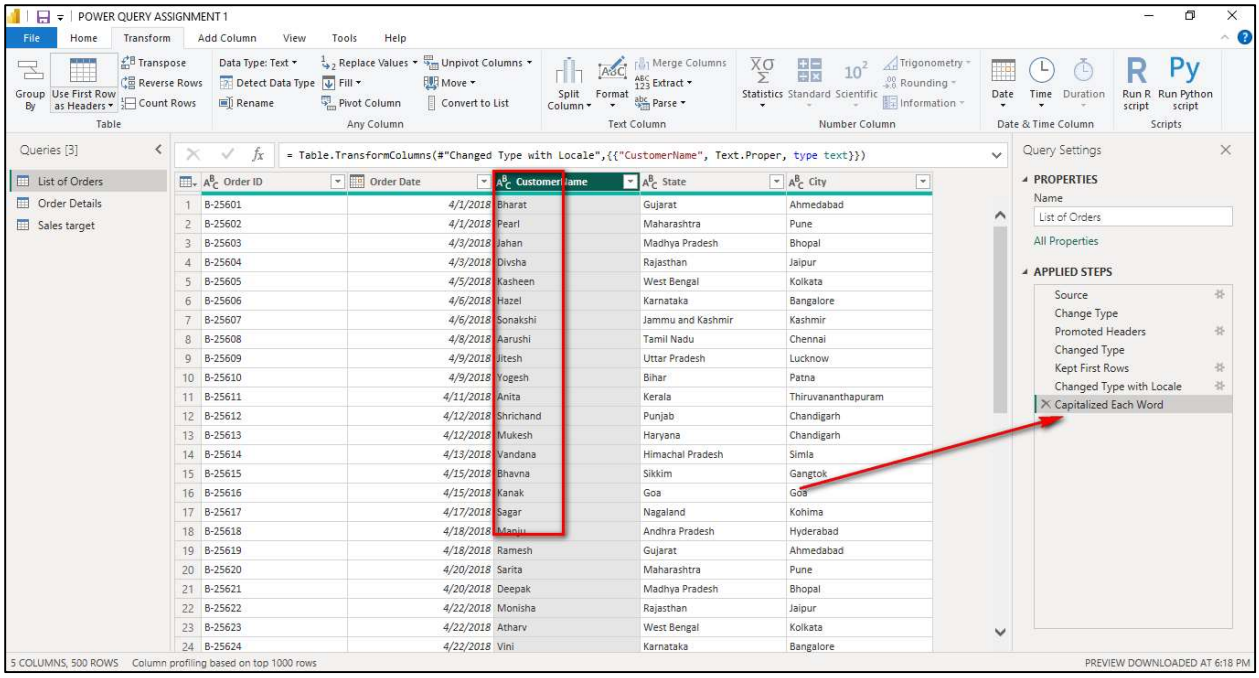
Target:



3. Text Formatting:

○ Format the Customer Name column to proper case to ensure consistent capitalization.

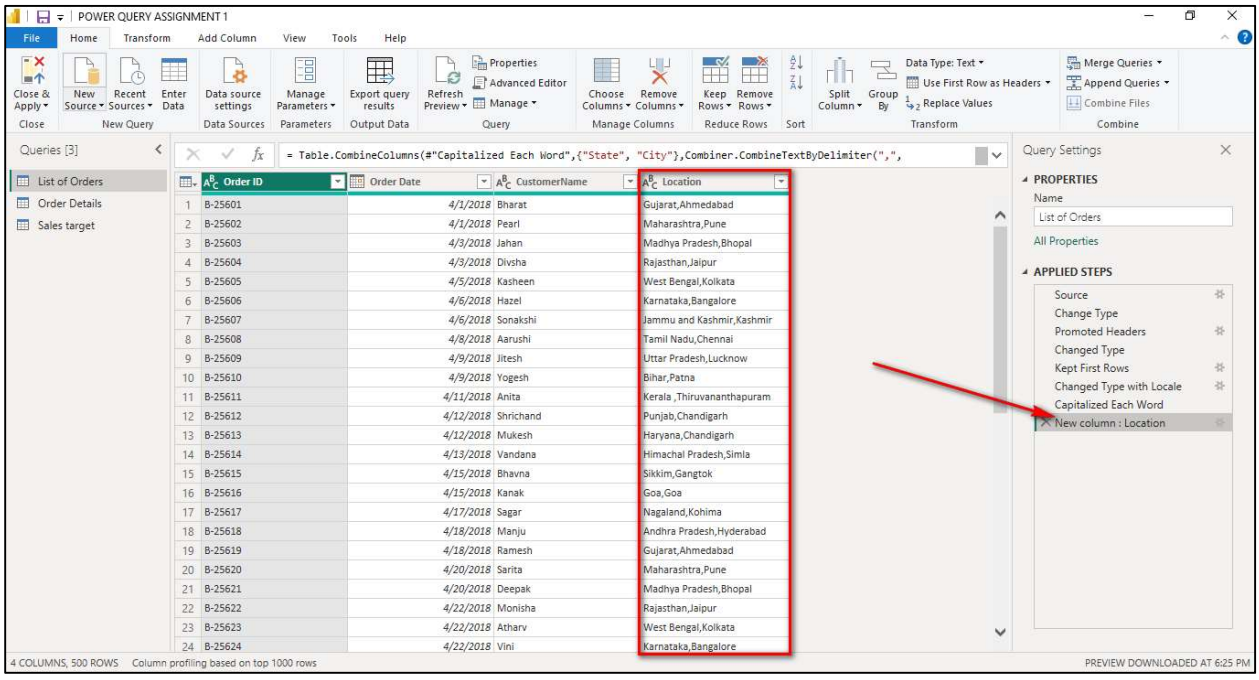




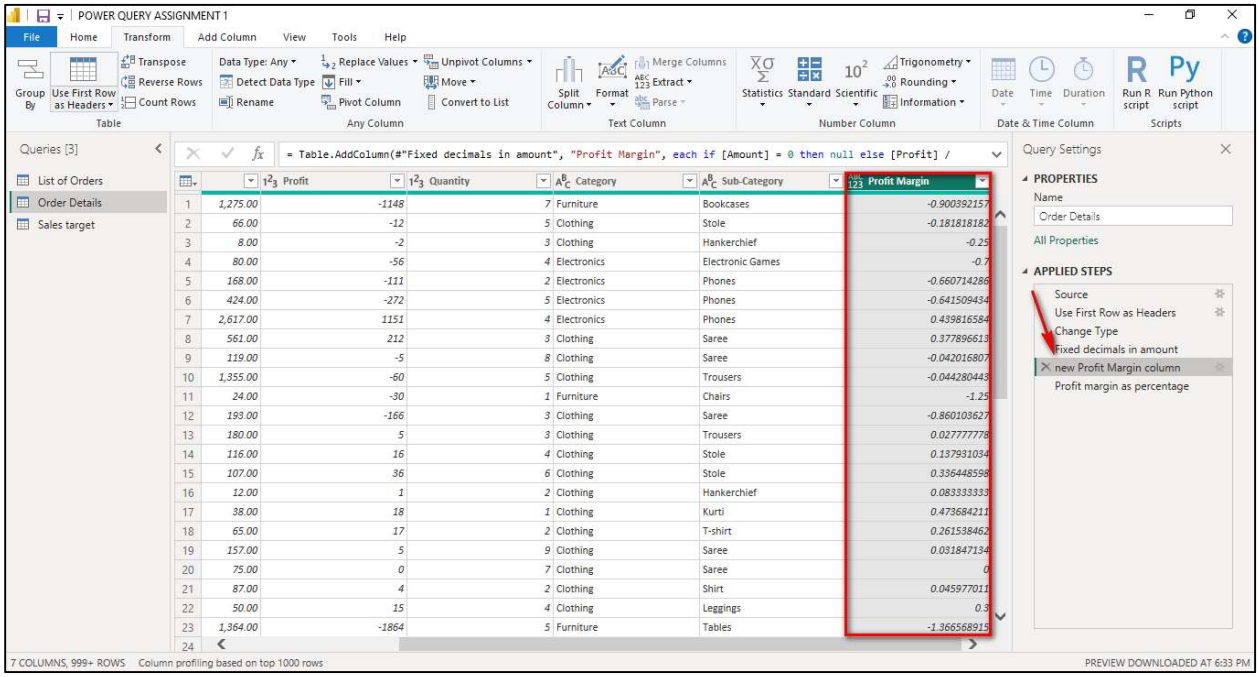
4. Column Creation:

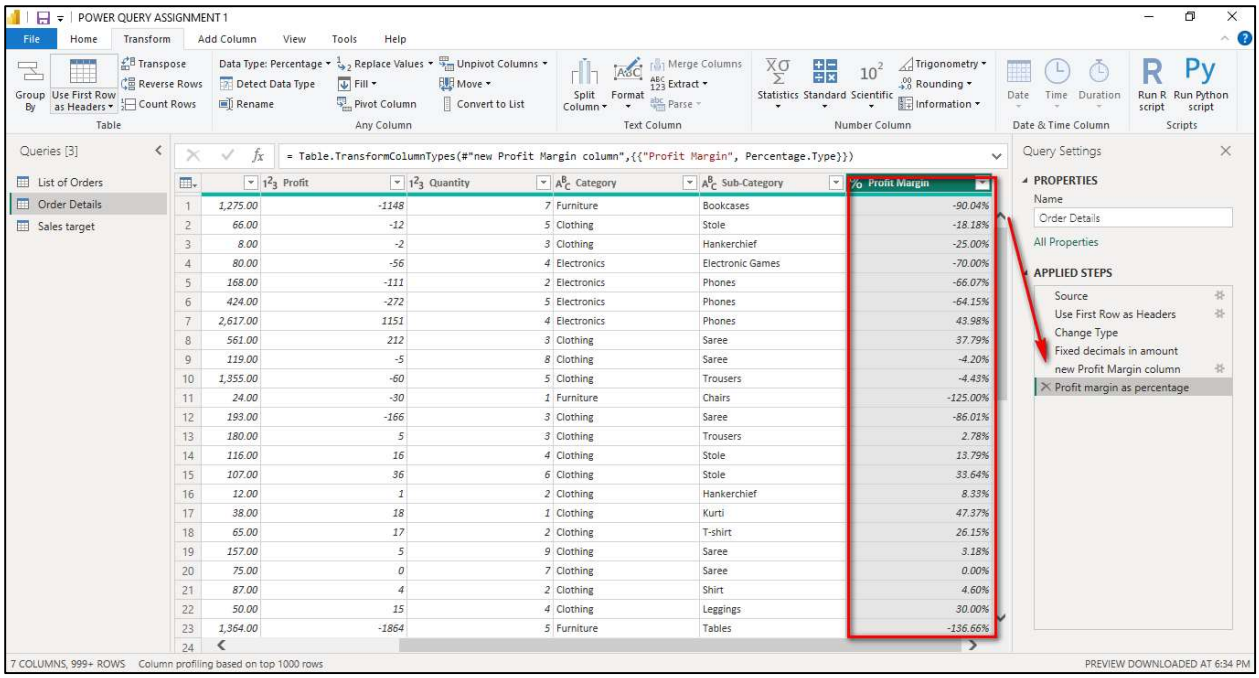
○ Merge the City and State columns to create a new column named Location in the format:

City, State



○ Create a custom column named Profit Margin calculated as: Profit / Amount (expressed as a percentage)





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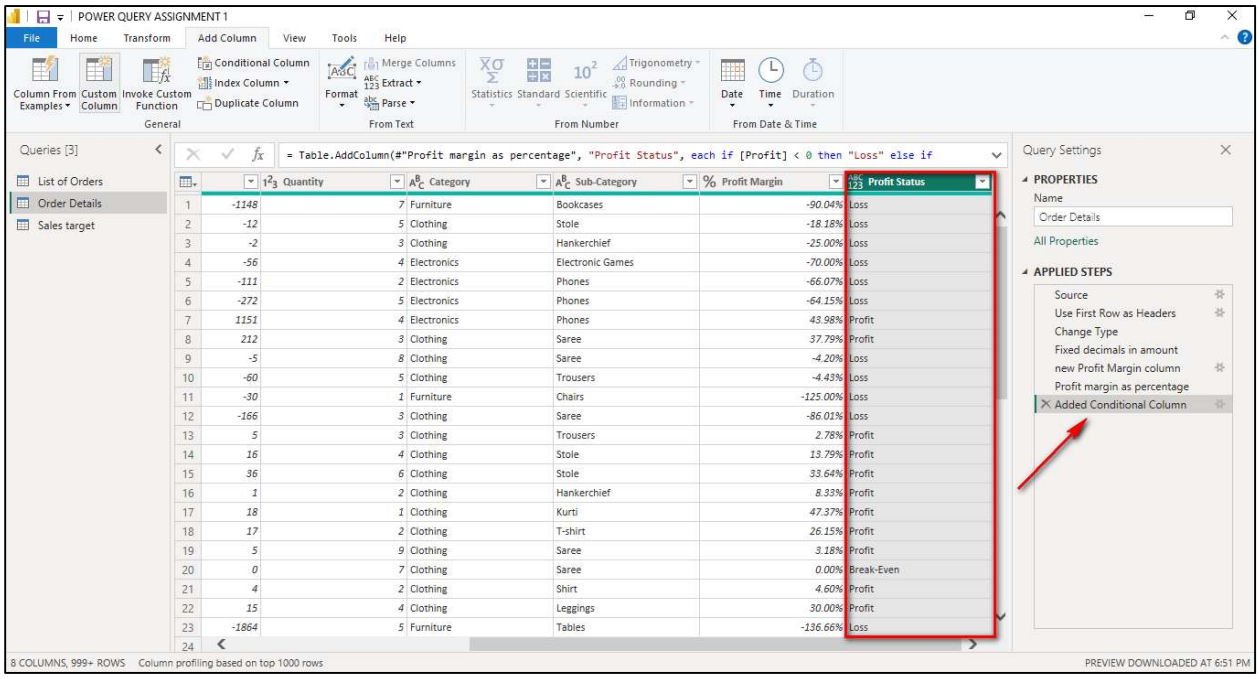
5. Conditional Column:

○ Add a conditional column named Profit Status based on:

■ Profit < 0 → Loss

■ Profit = 0 → Break-Even

■ Profit > 0 → Profit

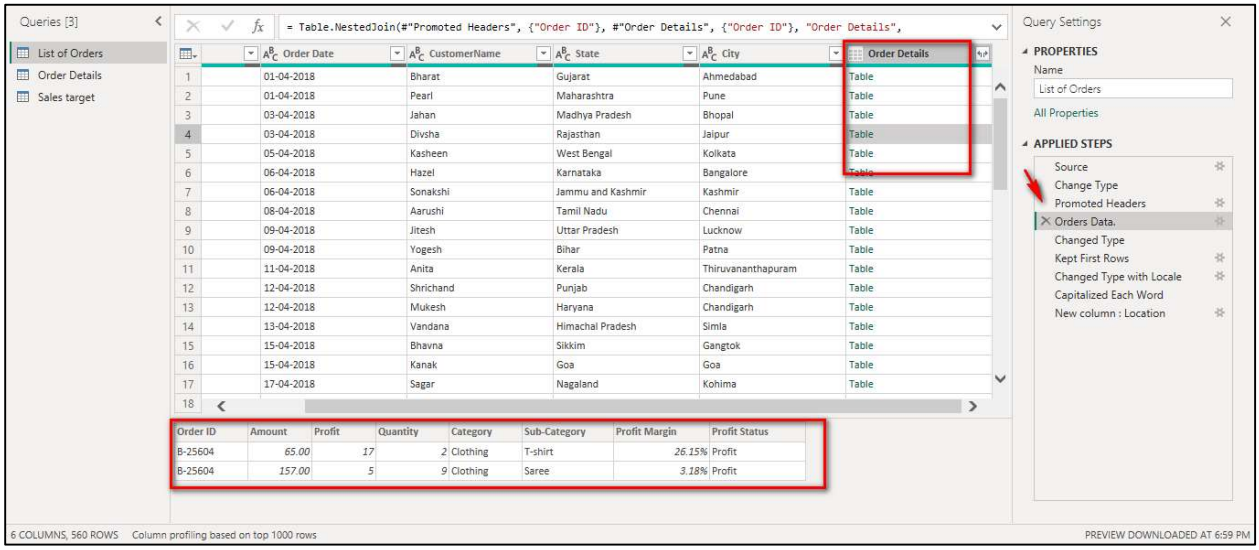


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6. Merging Data (Joins)

○ Merge the List of Orders and Order Details tables using Order ID.

○ Name the merged output table as Orders Data.



7. Handling Missing Data & Duplicate Data

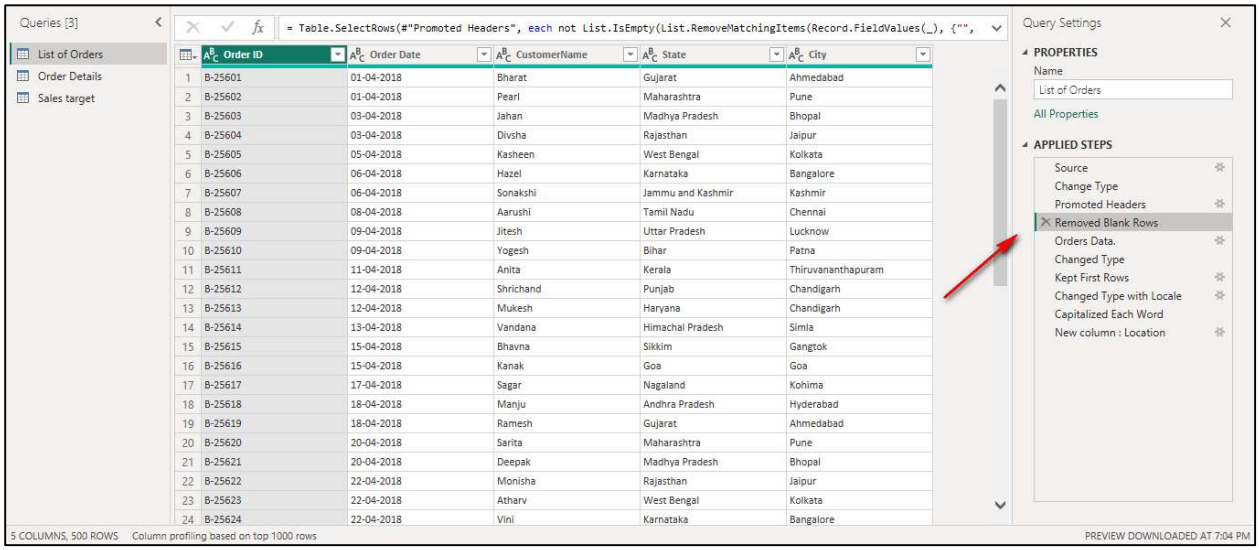
● Identify missing values across all tables.

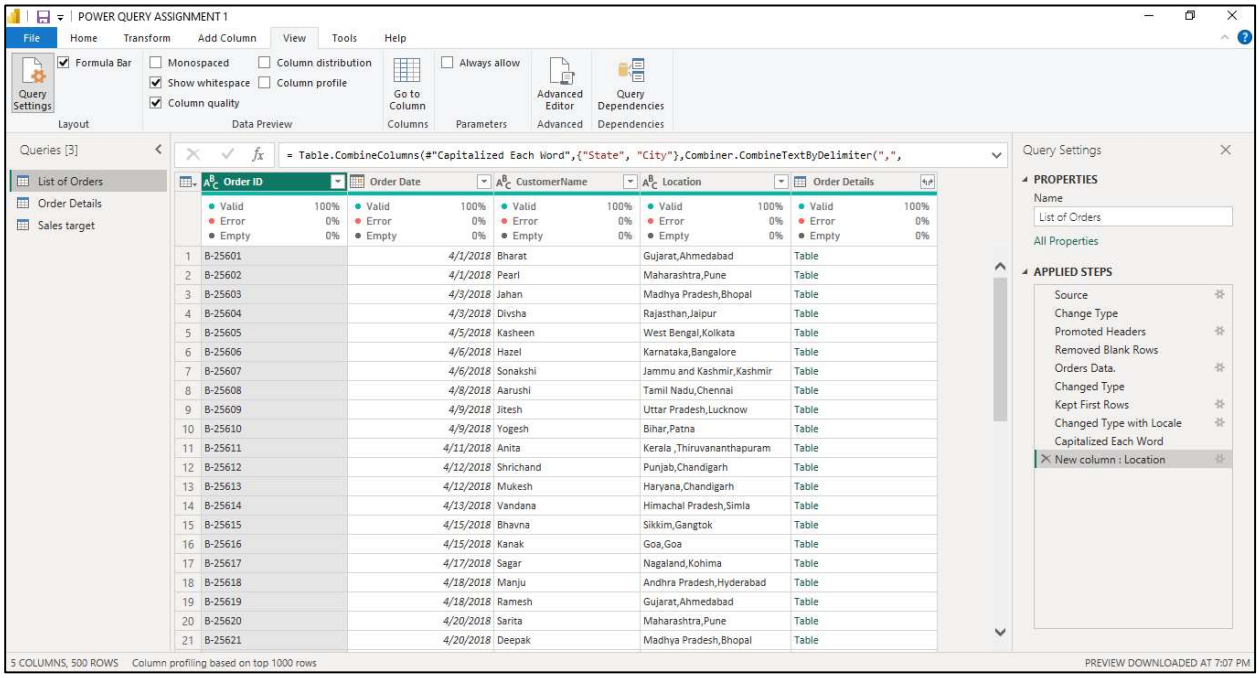
● Define and apply appropriate strategies such as:

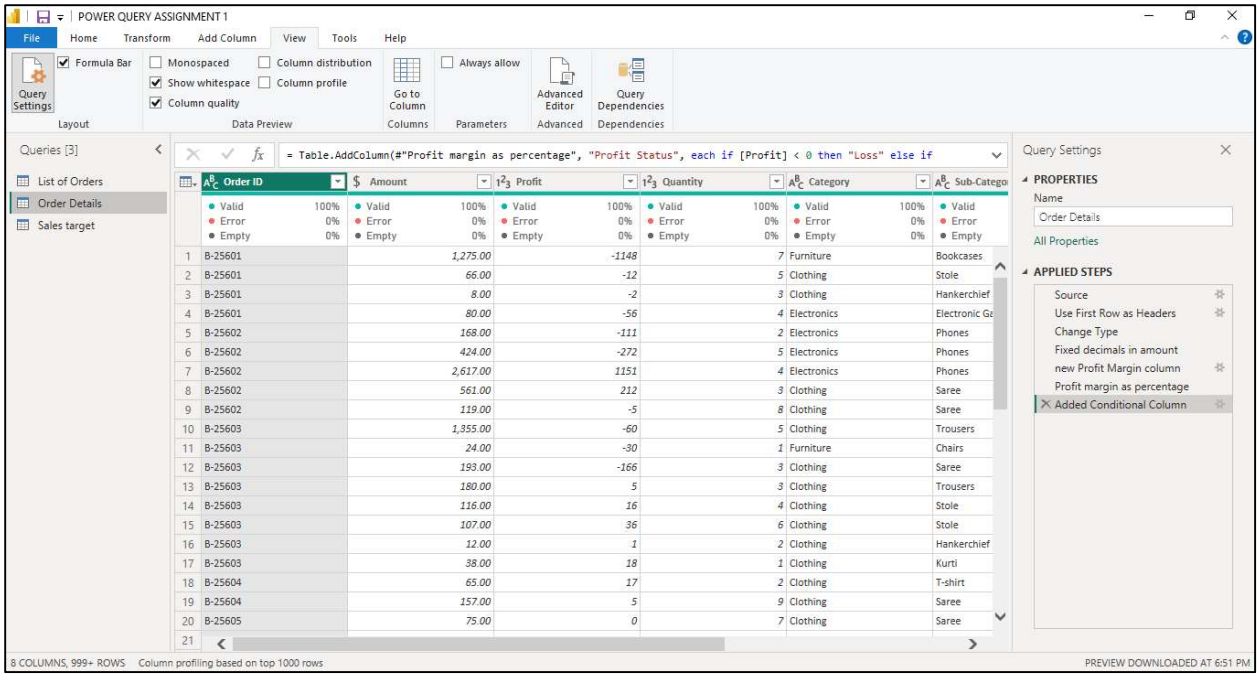
○ Replacing missing values

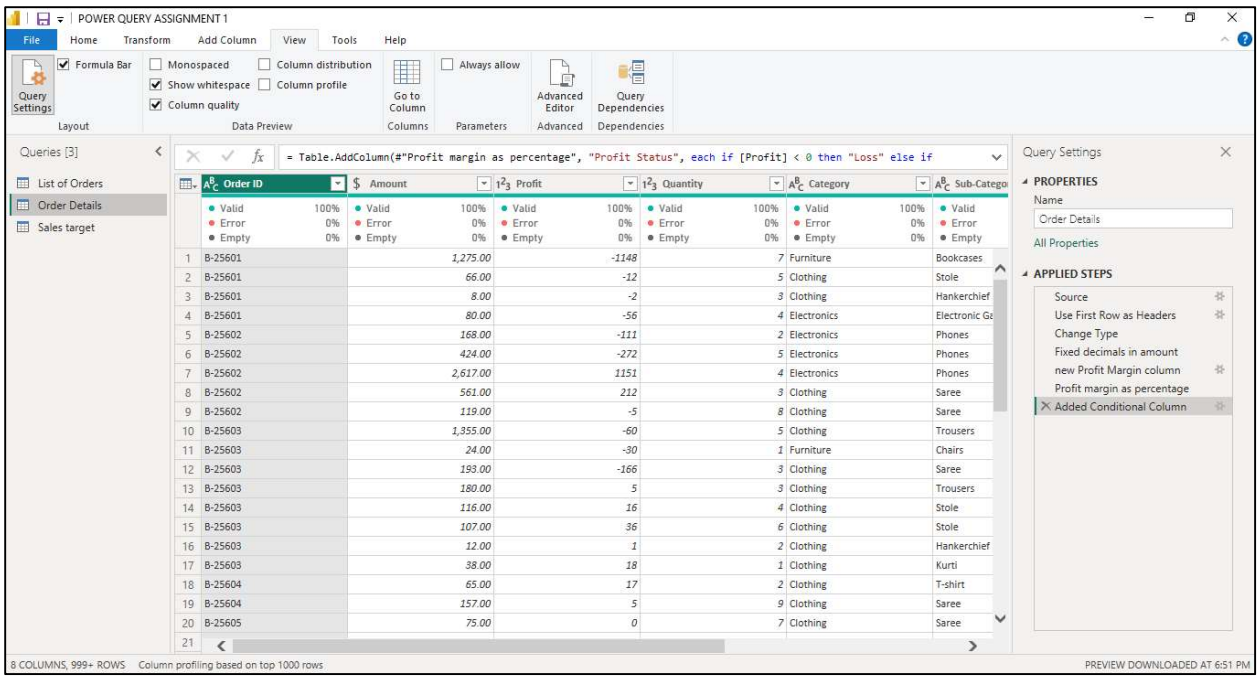
○ Removing records

○ Retaining nulls with justification





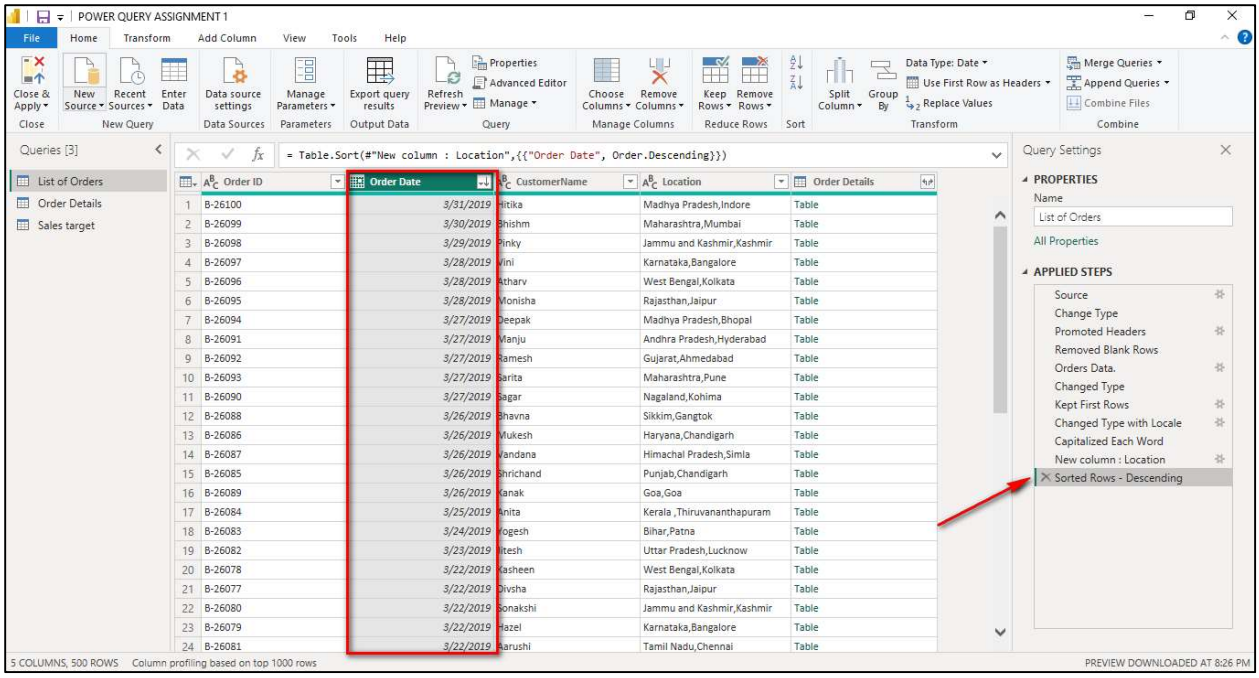




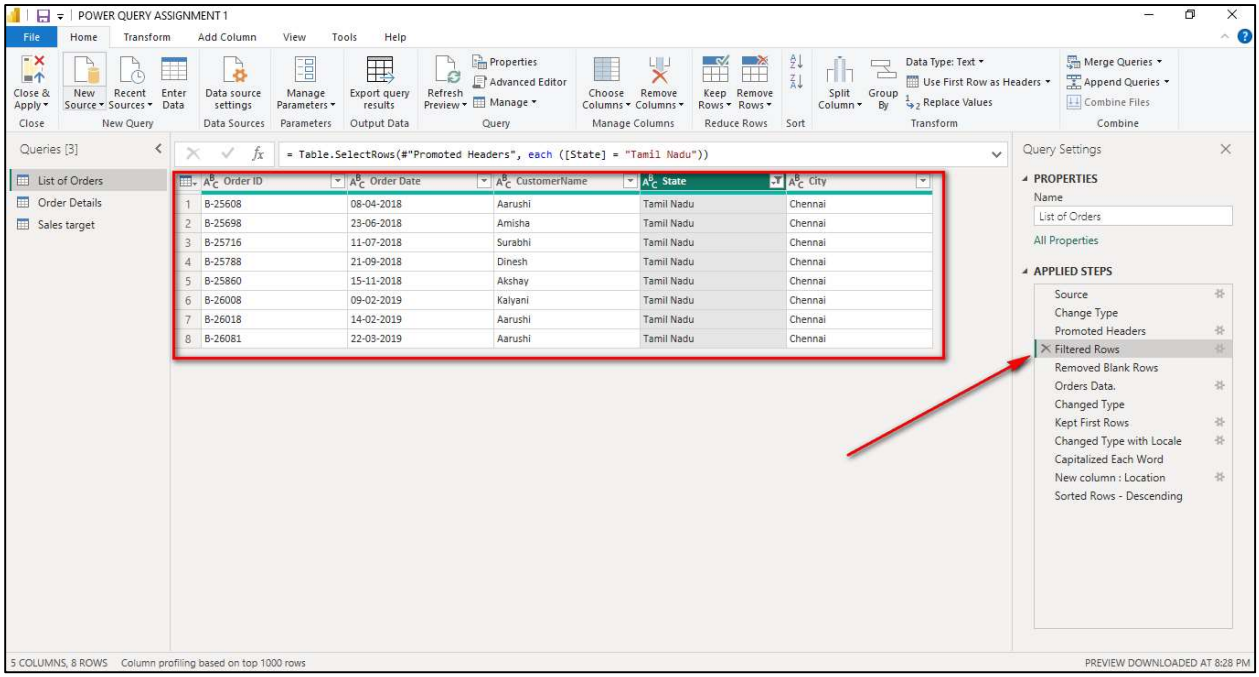
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8. Sorting and Filtering Data

- Sort records in the Orders Data table by Order Date in descending order.



● Apply filters to analyze orders from a specific state (e.g., Tamil Nadu).

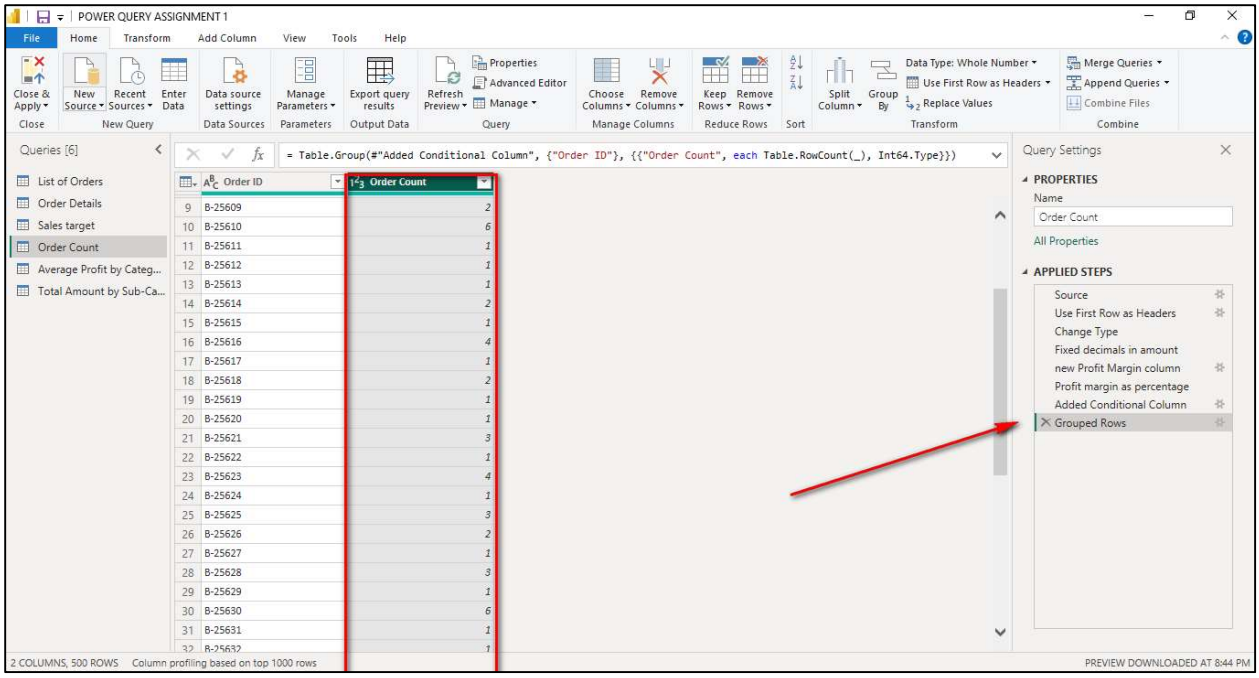


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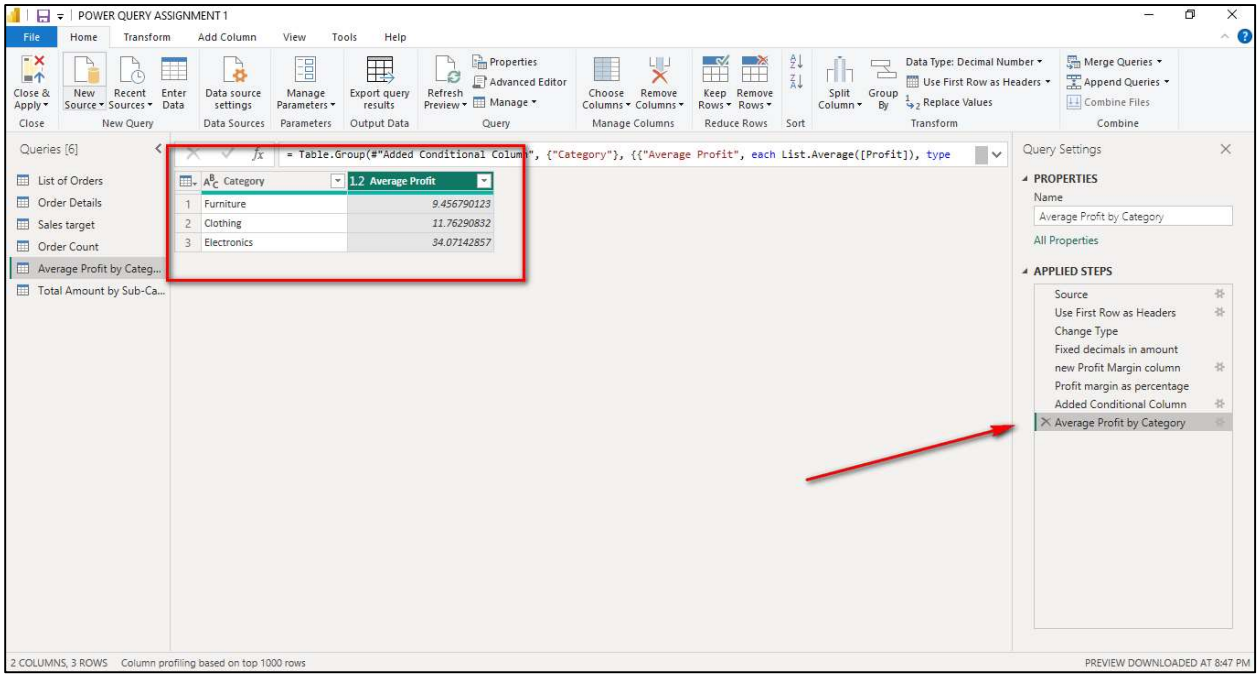
9. Grouping and Aggregating Data

● Duplicate the Order Details table and perform the following:

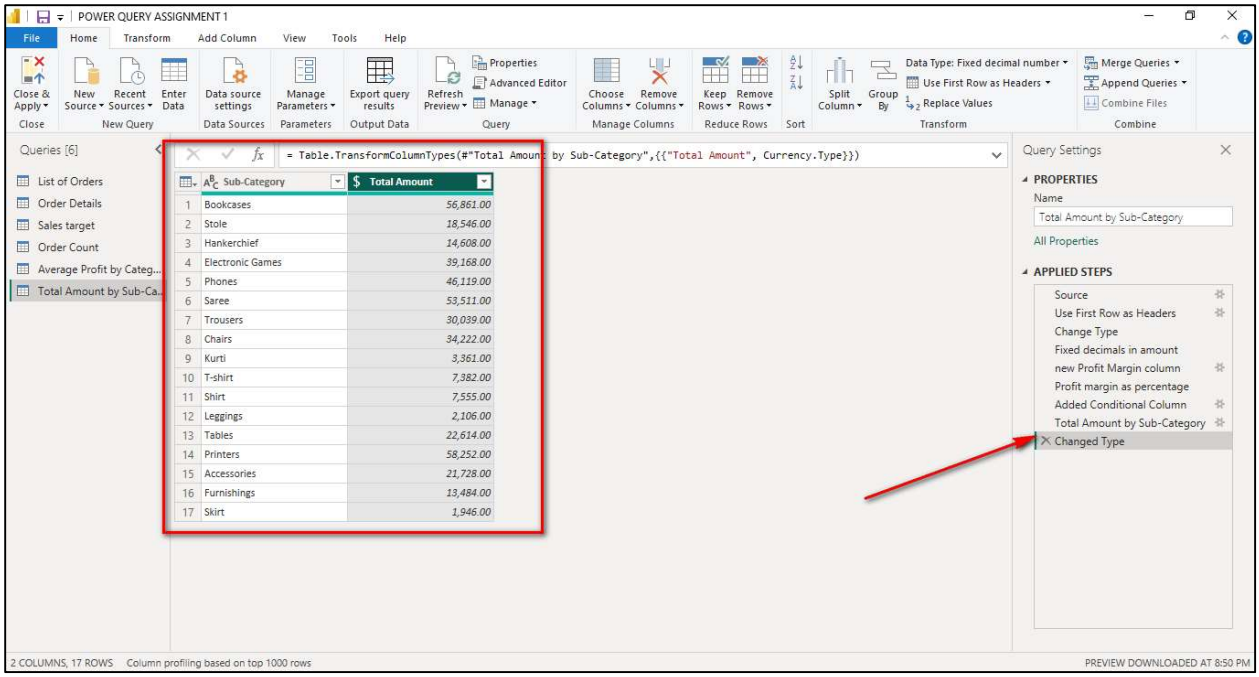
○ Count of Order ID



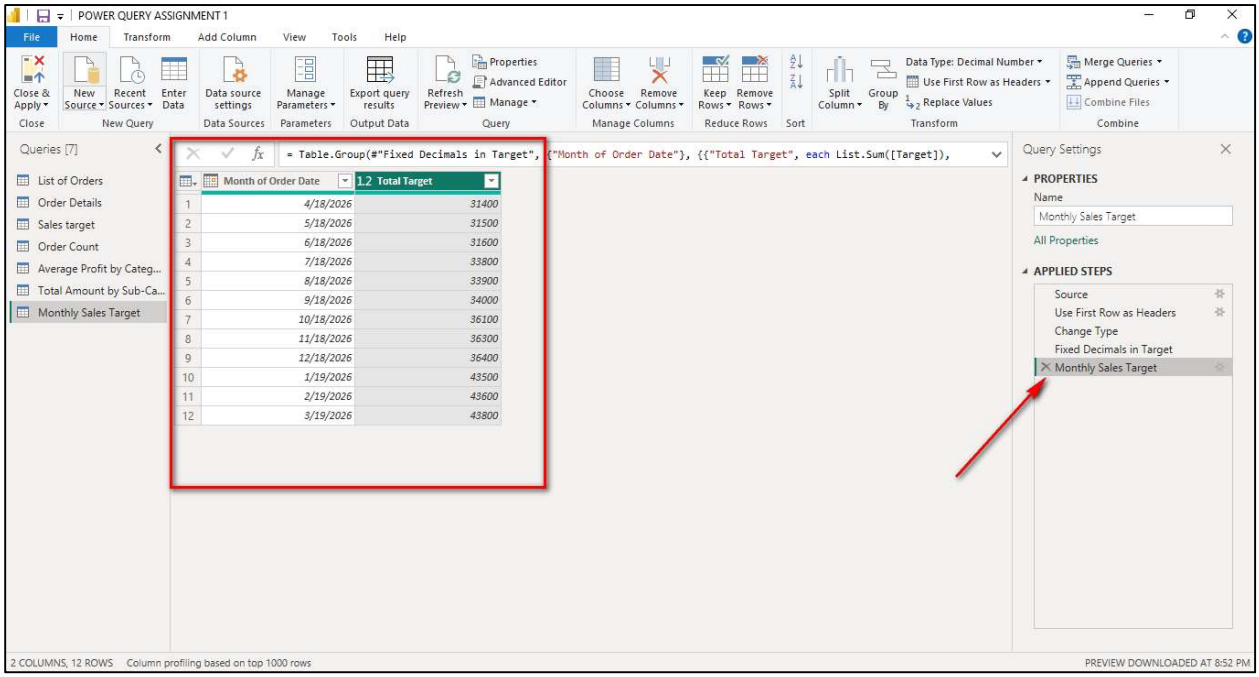
○ Average Profit by Category



○ Total Amount by Sub-Category



● Duplicate the Sales Target table and aggregate the total Target by Month of order date.



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10. Data Modeling:

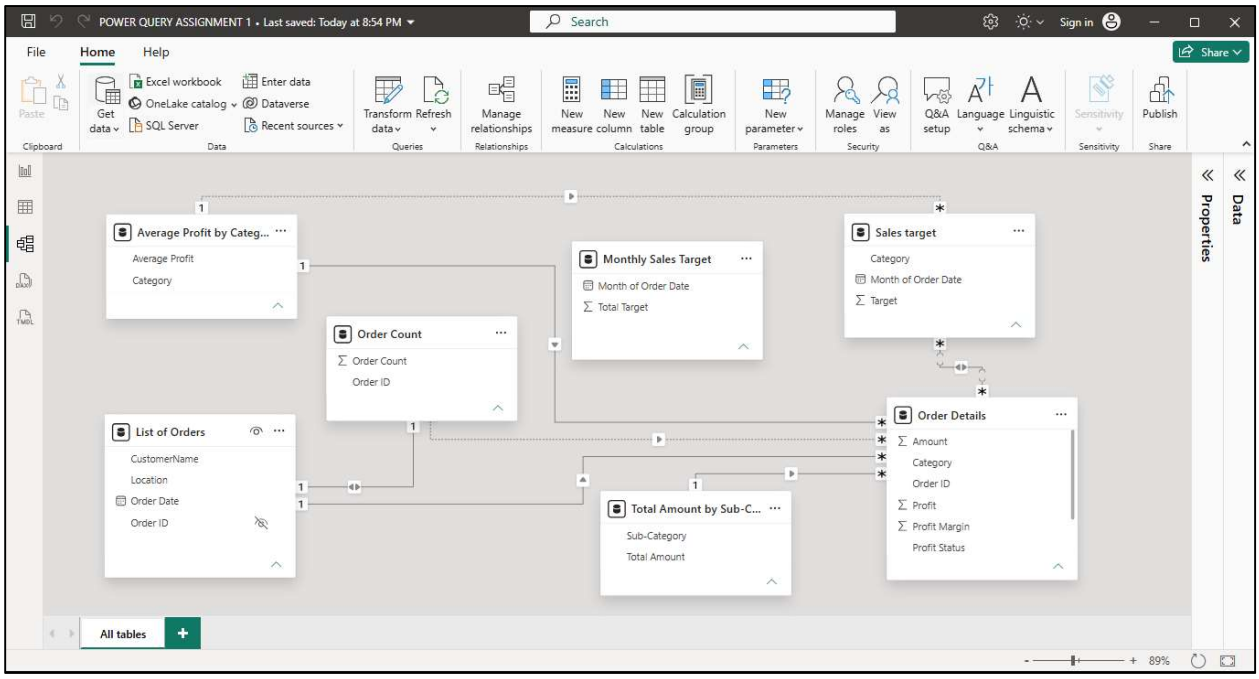
● Establish a relationship between:

○ List of Orders and Order Details using Order ID

● Establish a relationship between:

○ Order Details and Sales Target using Category

2. Use Manage Relationships to ensure relationships are active and correctly defined.



Brief Explanation of Transformation and Modeling Logic

1. Data Transformation Logic (Power Query)

Data transformation was performed to clean, organize and prepare the raw data for analysis.

1. Row Filtering: Kept only the first 500 rows in the List of Orders table.
2. Data Type Conversion: Changed Amount and Target columns to Fixed Decimal Number.
3. Text Formatting: Converted Customer Name into Proper Case for consistent capitalization.
4. Column Creation: Merged City and State to create a new Location column.
5. Custom Column: Created Profit Margin using Profit divided by Amount.
6. Conditional Column: Created Profit Status to classify records as Profit, Loss or Break-Even.

7. Data Merging: Merged List of Orders and Order Details using Order ID to create Orders Data.
8. Missing Data Handling: Removed completely blank rows and checked all tables for null values.
9. Duplicate Handling: Retained repeated Order IDs because they represent multiple product entries.
10. Sorting and Filtering: Sorted orders by Order Date in descending order and filtered Tamil Nadu orders.
11. Grouping and Aggregation: Created summary tables for Order Count, Average Profit by Category, Total Amount by Sub-Category and Monthly Sales Target.

2. Data Modeling Logic (Power BI)

Data modeling was performed to connect tables and enable analysis across different business categories.

Relationship	Key column	Relationship type
List of Orders → Order Details	Order ID	One-to-Many (1:*)
Order Details → Sales Target	Category	Many-to-Many (:)

Modeling steps:

1. Connected List of Orders and Order Details using Order ID.
2. Connected Order Details and Sales Target using Category.
3. Enabled both relationships.
4. Verified the relationships using Manage Relationships.
5. Used the connected tables to support sales, profit and target analysis.

3. Overall Purpose

- Transformation: Improves data quality and prepares raw data for analysis.
- Modeling: Connects related tables and supports meaningful business analysis.
- Analysis: Helps understand sales performance, profitability, category performance and target achievement.

Conclusion: Power Query was used to clean and transform the raw data, while Power BI data modeling was used to establish relationships between tables. This prepared the dataset for accurate business analysis and reporting.